

1.GCC

```
hecker@Windows: ~/Project_Os/Week_5
hecker@Windows:~/Project_Os/Week_5$ cat > main.c << 'EOF'
#include <stdio.h>

int main() {
    printf("Hello World\n");
    return 0;
}
EOF
hecker@Windows:~/Project_Os/Week_5$ gcc -Wall -g main.c -o main
hecker@Windows:~/Project_Os/Week_5$ ./main
Hello World
hecker@Windows:~/Project_Os/Week_5$ gcc -c main.c
hecker@Windows:~/Project_Os/Week_5$ gcc main.o -o myprogram
hecker@Windows:~/Project_Os/Week_5$ ./myprogram
Hello World
hecker@Windows:~/Project_Os/Week_5$ ldd myprogram
        linux-vdso.so.1 (0x0000798b22726000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x0000798b22400000)
        /lib64/ld-linux-x86-64.so.2 (0x0000798b22728000)
hecker@Windows:~/Project_Os/Week_5$
```

2.Build Helper Script

```
hecker@Windows: ~/Project_Os/Week_5
hecker@Windows:~/Project_Os/Week_5$ cat > build.sh << 'EOF'
#!/bin/bash

set -e
set -o pipefail

mkdir -p build bin

echo "Cleaning..."
rm -rf build/* bin/*

echo "Compiling..."
gcc -c main.c -o build/main.o

echo "Linking..."
gcc build/main.o -o bin/app

echo "Build Success"
EOF
hecker@Windows:~/Project_Os/Week_5$ chmod +x build.sh
hecker@Windows:~/Project_Os/Week_5$ ./build.sh
Cleaning...
Compiling...
Linking...
Build Success
hecker@Windows:~/Project_Os/Week_5$ ./bin/app
Hello World
hecker@Windows:~/Project_Os/Week_5$
```

3.Automated Backup Script

```
hecker@Windows: ~/Project_Os/Week_5
#!/bin/bash

if [ -z "$1" ]; then
    echo "Usage: $0 <directory>"
    exit 1
fi

if [ ! -d "$1" ]; then
    echo "Invalid directory"
    exit 1
fi

timestamp=$(date +%Y%m%d_%H%M%S)
tar -czf backup_${timestamp}.tar.gz "$1"

echo "Backup created: backup_${timestamp}.tar.gz"
EOF
hecker@Windows:~/Project_Os/Week_5$ chmod +x backup.sh
hecker@Windows:~/Project_Os/Week_5$ mkdir data
hecker@Windows:~/Project_Os/Week_5$ echo "test" > data/file.txt
hecker@Windows:~/Project_Os/Week_5$ ./backup.sh data
Backup created: backup_20260330_221239.tar.gz
hecker@Windows:~/Project_Os/Week_5$ ls
backup_20260330_221239.tar.gz  backup.sh  bin  build  build.sh  data  main  main.c  main.o  myprogram
hecker@Windows:~/Project_Os/Week_5$ tar -tzf backup_*.tar.gz
data/
data/file.txt
hecker@Windows:~/Project_Os/Week_5$
```